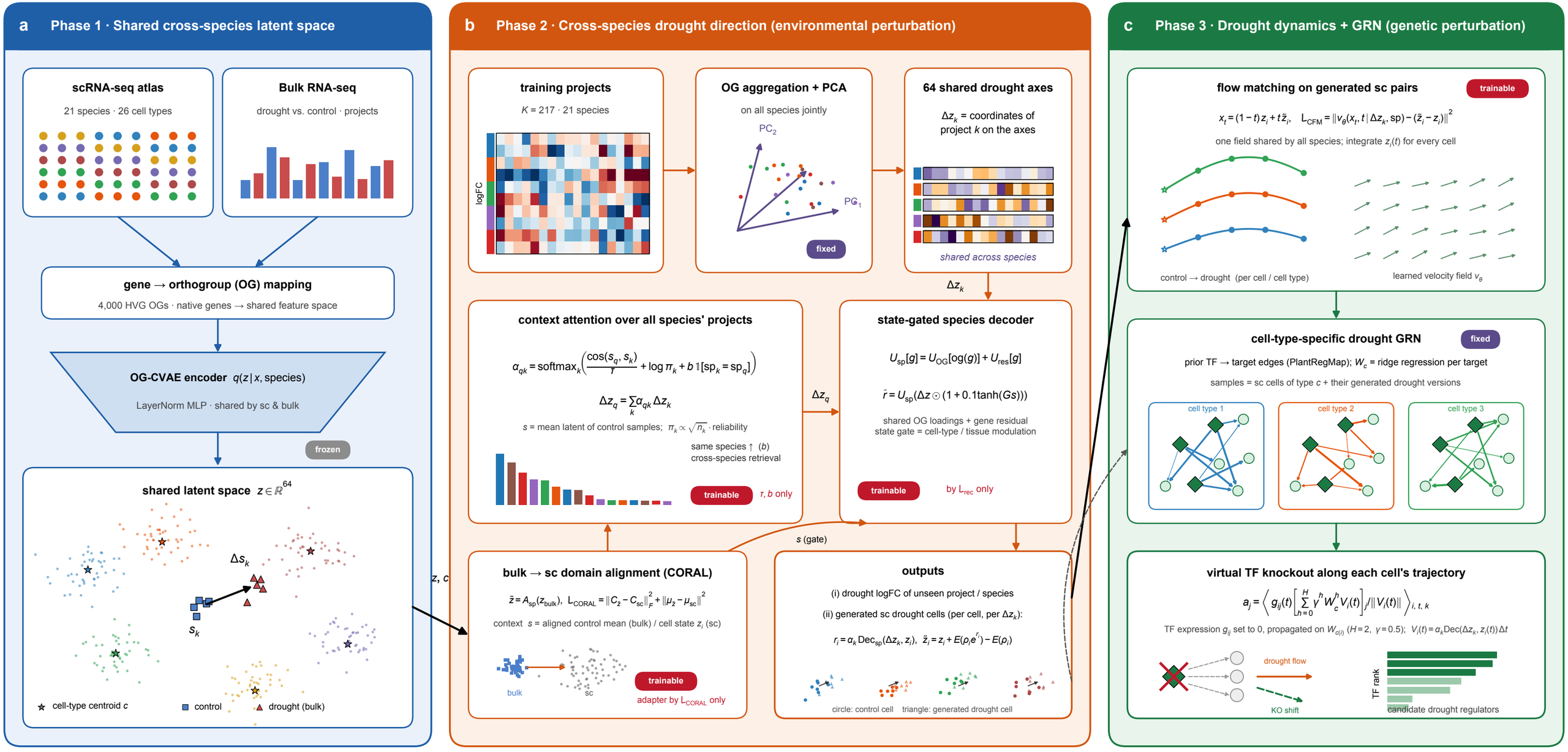

a Phase 1 · Shared cross-species latent space
scRNA-seq atlas
21 species · 26 cell types
Bulk RNA-seq
drought vs. control · projects
gene → orthogroup (OG) mapping
4,000 HVG OGs · native genes → shared feature space
OG-CVAE encoder q(z | x, species)
LayerNorm MLP · shared by sc & bulk
frozen
shared latent space z ∈ ℝ^64
Δs_k
s_k
cell-type centroid c
control
drought (bulk)
z, c
b Phase 2 · Cross-species drought direction (environmental perturbation)
training projects
K = 217 · 21 species
logFC
OG aggregation + PCA
on all species jointly
PC2
PC1
fixed
64 shared drought axes
Δz_k = coordinates of project k on the axes
shared across species
Δz_k
context attention over all species' projects
α_qk = softmax_k(cos(s_q, s_k)/τ + log π_k + b 𝟙[sp_k = sp_q])
Δz_q = Σ_k α_qk Δz_k
s = mean latent of control samples; π_k ∝ √n_k · reliability
same species ↑ (b)
cross-species retrieval
trainable
τ, b only
Δz_q
state-gated species decoder
U_sp[g] = U_OG[log(g)] + U_res[g]
r̂ = U_sp(Δz ⊙ (1 + 0.1tanh(Gs)))
shared OG loadings + gene residual
state gate = cell-type / tissue modulation
trainable
by L_rec only
s (gate)
bulk → sc domain alignment (CORAL)
z̃ = A_sp(z_bulk), L_CORAL = ‖C_z̃ − C_sc‖²_F + ‖μ_z̃ − μ_sc‖²
context s = aligned control mean (bulk) / cell state z_i (sc)
bulk
sc
trainable
adapter by L_CORAL only
outputs
(i) drought logFC of unseen project / species
(ii) generated sc drought cells (per cell, per Δz_k):
r_i = α_k Dec_sp(Δz_k, z_i), z̃_i = z_i + E(ρ_i e^{r_i}) − E(ρ_i)
circle: control cell
triangle: generated drought cell
c Phase 3 · Drought dynamics + GRN (genetic perturbation)
flow matching on generated sc pairs
trainable
x_t = (1 − t) z_i + t z̃_i, L_CFM = ‖v_θ(x_t, t | Δz_k, sp) − (z̃_i − z_i)‖²
one field shared by all species; integrate z_i(t) for every cell
control → drought (per cell / cell type)
learned velocity field v_θ
cell-type-specific drought GRN
fixed
prior TF → target edges (PlantRegMap); W_c = ridge regression per target
samples = sc cells of type c + their generated drought versions
cell type 1
cell type 2
cell type 3
virtual TF knockout along each cell's trajectory
a_j = ⟨ g_ij(t) [Σ_{h=0}^{H} γ^h W_c^h V_i(t)]_j / ‖V_i(t)‖ ⟩_{i, t, k}
TF expression g_ij set to 0, propagated on W_c(i) (H = 2, γ = 0.5); V_i(t) = α_k Dec(Δz_k, z_i(t)) Δt
drought flow
KO shift
TF rank
candidate drought regulators
Training objectives (only what each loss trains)
L_rec: each training project's logFC from its own Δz_k → decoder (U_OG, U_res, G)
L_LOO: project k from the other projects (decoder detached) → τ, b only
L_CORAL → adapter only; L_CFM → velocity field v_θ; GRN W_c: closed-form ridge
Results vs. best baseline (3 seeds)
Phase 1 cell-type F1 within / cross-species 0.766 / 0.758 (scANVI 0.691 / 0.502)
Phase 2 drought-response Spearman 0.311 (B1 0.262); cross-species 0.212 (B2 0.173)
Phase 3 known drought-TF AUROC / AUPRC 0.721 / 0.618 (DE 0.627 / 0.548)
Legend
frozen
Phase 1 weights
fixed
PCA / ridge
trainable
gradient-trained
species